
**Security Bid/Ask Dynamics with Discreteness and Clustering:
Simple Strategies for Modeling and Estimation**

Joel Hasbrouck

First Draft: September 22, 1997

This Draft: January 20, 1998

Preliminary Draft
Not for Quotation
Comments Welcome

Professor of Finance
Stern School of Business
New York University
Suite 9-190 Mail Code 0268
44 West Fourth St.
New York, NY 10012-1126

Tel: (212) 998-0310

Fax: (212) 995-4901

E-mail: jhasbrou@stern.nyu.edu

Web: <http://www.stern.nyu.edu/~jhasbrou>

Olsen Associates provided the data used in this study. The Vincent Ross Institute of Accounting at the Stern School provided financial support. For suggestions and comments on this work, I am grateful to Charles Goodhart, Jim Hamilton, Michael Melvin, Neil Shephard and workshop participants at Columbia and the University of Pennsylvania. This acknowledgement should not be construed as implying endorsement. All errors are my own responsibility.

**Security Bid/Ask Dynamics with Discreteness and Clustering:
Simple Strategies for Modeling and Estimation**

Abstract

The short-term movements of a security price reflect the latent efficient price (conditional expectation of terminal value) and also components arising from the trading mechanism itself. Observed bid and ask quotes are but rough signals of these unobserved quantities. The bid and ask quotes in the \$/DM market considered here, for example, are discrete, with a tick size that is not trivial relative to the spread. Furthermore, the distribution of these quotes is clustered, with a greater-than-expected incidence of five-tick multiples. This paper suggests a simple framework for handling discrete, clustered quotes. Despite the simplicity of the model, estimation by traditional (likelihood or moment) methods is difficult. As an alternative, the paper implements a Gibbs sampler approach that proves to be quick and effective. This strategy opens the door for the investigation of a broad class of structural microstructure models.

1. Introduction.

Security price dynamics over short horizons are frequently viewed as reflecting long run or permanent informational components and also transient components arising from the trading mechanism. Whether one regards these latter, microstructure components as chief concerns or nuisance effects, cursory consideration of their magnitude suggests that they be addressed in some fashion in the modeling process. To this end, this paper proposes a simple statistical model for bid and ask quote dynamics that incorporates several key microstructure effects: a stochastic cost of market-making, discreteness and clustering. The model is especially amenable to estimation via Markov chain Monte Carlo methods. An application to bid and ask quotes in the U.S. Dollar/Deutschemark market provides an illustration.

The model may be summarized as follows. The permanent price component common to the observed bid and ask quotes follows a random walk. The dealer posting a bid or ask quote is subject to a stochastic trading cost that encompasses clearing, asymmetric information and inventory (position management) costs specific to the next trade and some allocation of fixed costs. These costs imply ask and bid reservation prices (the permanent component plus and minus the trading cost), which map in turn onto the posted discrete quotes via a stochastic rounding process.

The rounding process is governed by a stochastic tick size. Although there is in most markets an explicit minimum tick established by regulation or technical limitation, actual market practice often suggests an implicit conventional tick size that is somewhat larger. In the DM/\$ market, for example, the minimum tick available on the computer systems is 0.0001 DM/\$. Nevertheless, bid and ask quotes exhibit a tendency to concentrate on multiples of 0.0005, an oft-found characteristic in security markets that is known as clustering. In the present model, the implicit tick size is viewed as a Bernoulli random variable. Given the implicit tick size, the rounding is asymmetric. The posted bid quote is the permanent component less the cost of trading rounded down to the next available (implicit) tick. Similarly, the posted ask quote is the permanent component plus the trading cost rounded up.

Considered separately, the features of the model figure prominently in the microstructure literature. There is strong theoretical and empirical evidence regarding the fixed cost, information and inventory components of the dealer's trading cost. O'Hara (1995) surveys theoretical models. Hasbrouck (1996) reviews empirical approaches to measuring and resolving these components in a linear covariance-stationary time series framework.

Discreteness of security prices became an important empirical consideration when option valuation applications gave rise to a need for estimates of stock price volatility. The earliest papers view transaction prices as symmetrically rounded random walks. Later work incorporates more realistic and complete microstructure effects.¹ Although statistical models incorporating discreteness have become progressively more refined, however, the normative analysis of economic behavior with a discrete price set is not yet fully developed.² As noted below, the rounding scheme used in the present paper is consistent with Bertrand competition among dealers in some (but not all) market structures.

The economic principles surrounding implicit tick conventions are even less clear. The tendency of bid, ask and transaction prices to cluster at "natural" multiples of the minimum tick is a fairly well established empirical regularity. Writing prior to the recent switch to sixteenths Harris (1991) notes that for U.S. equity markets,

¹ Gottlieb and Kalay (1985) and Ball (1988) advocate the rounded-random-walk paradigm. The analyses of Dravid (1991) and Harris (1990) incorporate a bid/ask spread in a covariance-stationary method-of-moments framework. Probit models have been estimated for discrete transaction price changes (Hausman, Lo and MacKinlay (1992)) and spreads (Bollerslev and Melvin (1994)). Closest in approach to the present paper are the latent-variable state-space models used by Glosten and Harris (1988) and Hasbrouck (1997). Hasbrouck (1997) reviews these and related developments.

² Economic analyses of the effect of the tick size on agent behavior and/or the optimal tick size include Ahn, Cao and Choe (1996), Angel (1994), Anshuman and Kalay (1994), Bernhardt and Hughson (1996), Brown, Laux and Schacter (1991), Chordia and Subrahmanyam (1995), Cordella and Foucault (1996), Glosten (1994), Harris (1990, 1991).

Stock [transaction] prices cluster on round fractions. Integers are more common than halves; halves are more common than odd quarters; odd quarters are more common than odd eighths; other fractions are rarely observed. The phenomenon is remarkably persistent through time and across stocks.

Harris (1991, 1994, and 1997) suggests that an implicit supra-minimum tick convention arises from traders seeking to minimize negotiation costs, to avoid extended rounds of bargaining over amounts of diminishing importance. Christie and Schultz (1994a, 1994b) find dramatic quote clustering in the U.S. over-the-counter NASDAQ equity market (prior to recent reforms). They suggest that the clustering reflected a collusive coordination mechanism, a hypothesis to which subsequently subpoenaed trading room tapes lend considerable support.³

Despite the simplicity of the present model, estimation via direct computation of the likelihood function is difficult. There are three latent random state variables (the permanent price component, the cost of trading and the implicit tick size), and the update at each time point therefore involves a three-fold integration. Hasbrouck (1997) proceeds in this manner using a similar model. As an alternative to likelihood estimation, this paper employs a Markov chain Monte Carlo (MCMC) technique, a Gibbs sampler that characterizes the distribution of model parameters as the marginal of a simulated distribution that is joint over the parameters, latent data and observed data. Gibbs samplers have been used to estimate stochastic volatility models and implement Bayesian analyses of various sorts. This paper is an attempt to extend these analyses into the realm of market microstructure.⁴

³ The investigation report of the U.S. Securities and Exchange Commission (U.S. S.E.C. (1996)) documented numerous instances wherein a dealer would call a “competitor” to complain about aggressive pricing. This direct evidence notwithstanding, various economic analyses address the question of what features of the market structure appear to support collusive equilibria. These and other aspects of the NASDAQ controversy are surveyed by Schwert (1997).

⁴ Excellent general introductions to MCMC analysis include Casella and George (1992), Chibb and Greenberg (1995), Gilks, Richardson and Spiegelhalter (1996), Smith and Gelfand (1992) and Tanner (1996). Chibb and Greenberg (1996) survey applications in econometrics. Jacquier, Polson and Rossi (1994) present an application to stochastic volatility modeling (see in addition

The paper closest to the present study is an MCMC analysis of a subcase of Hasbrouck's (1997) model due to Manrique and Shephard (1997b). The present model generalizes on Hasbrouck (1997) and Manrique and Shephard (1997b) in its incorporation of clustering, but it is more restricted in other respects.

Although the estimation here is cast as a Bayesian analysis, the Bayesian content of the analysis is low in the sense that noninformative priors are used throughout. The advantages of working in Bayesian framework are computational. With the availability of MCMC methods, it is often easier to estimate a model in a Bayesian framework than with likelihood methods. In many applications (including those in market microstructure), MCMC methods also possess an advantage in their relative ease of generalization. The computational requirements of a likelihood calculation are approximately exponential in the number of state variables, while those of an MCMC estimation are roughly linear. This lifting of the curse of dimensionality opens the door for models that incorporate endogenous trades and time-varying and stochastic parameters (such as volatility and clustering probabilities).

The paper is organized as follows. First, to motivate the model and provide an application, the next section characterizes in a preliminary fashion a sample of daily \$/DM exchange rate quotes. A simple model broadly consistent with the features of the data is presented in Section 3. Section 4 discusses estimation via traditional methods and the alternative MCMC strategy. Results for the \$/DM data are reported in Section 5. Section 6 sketches out extensions to the simple model. A brief summary concludes the paper in Section 7.

2. A preliminary look at the data.

As a representative series, this study examines bid and ask quotes in the U.S. Dollar/Deutschemark market. The structure is a continuous dealer market. Bid and ask quotes

the comments and response for this paper in the same volume). Kim, Shephard and Chibb (1998) discuss further aspects of stochastic volatility modeling. Key references in the MCMC analysis of state space models include Carter and Kohn (1994), de Jong and Shephard (1995) and Shepard and Pitt (1997).

are disseminated by Reuters (among other systems), but there is also an active interdealer market for which no public quotes are visible. Bessembinder (1994), Bollerslev and Melvin (1994), Lyons (1995), Peiers (1997) and Yao (1997) discuss the structure of this market and certain features of the data relevant to the present study.

The sample consists of bid and ask quotes prevailing on the Deutschemark as of noon (GMT) on regular U.K. business days in 1996, quoted in DM per U.S. dollar. The data are Reuters quotes from the Olsen Associates HFDF II data set. The frequency of this data set is hourly, but the series analyzed here is the extracted daily series (as of noon GMT), roughly the middle of the London trading day. Most of the features of present interest appear in this daily series and use of daily data avoids the need to consider intraday periodicities.

The Reuters quotes are indicative quotes entered by participating dealers. The bid and ask quotes used here are simply those most recently entered (as of noon GMT). Although they are the most timely quotes, they are not necessarily “best”, since the bid and ask (or both) might be dominated by prices submitted earlier that is still considered active by its issuer. These quotes are not, in any event, firm. Furthermore, they are often dominated by bids and asks available in the interdealer market (Lyons (1995)) and on electronic trading systems such as the Reuters D2000-2 system.⁵

Prices are quoted to four decimal places, e.g., 1.4123 DM/\$. The tick size imposed by the information systems is thus 0.0001 DM/\$. For convenience, all prices in this paper will be reported in ticks, e.g., 14,123.

Figure 1 depicts the distribution of the spread (the ask less the bid). The spread exhibits considerable variation, but is almost always ten ticks or less. This suggests that the resolution of the spread is generally less than ten percent of its magnitude. Even in light of the discreteness mandated by the tick size, however, the distribution of the spread can by no means be considered smooth. The distribution is sharply bimodal, with peaks at five and ten ticks. Bessembinder

⁵ I am indebted to Charles Goodhart and Michael Melvin for their comments on the limitations of the Reuters quotes. This acknowledgement should not be construed as implying, however, their approval or endorsement of the present interpretation or use of these quotes.

(1994) and Bollerslev and Melvin (1994) note spread clustering in the Reuters indicative quotes. Goodhart et al. (1996) note, however, that clustering is not observed in the D2000-2 quotes.

It might be hypothesized that this clustering simply reflects a bimodal cost structure, two regimes in which the costs of market making are near five and ten ticks. This hypothesis can be investigated by considering the distribution of bid and ask prices separately. Figure 2 plots the distribution of the last digit of the bid and ask prices. If the clustering of the spread were arising from clustering in the underlying costs, one would still expect to observe a flat distribution of price last-digits. Figure 2 suggests that this is not the case. The bid and ask prices considered separately exhibit considerable clustering.

Not only do we observe a higher-than-expected occurrence of five- and ten-tick bids and offers, but if the bid lies on a five-tick multiple, there is a higher probability that the ask also lies on a five-tick multiple. A simple classification test soundly rejects the null hypothesis of independence. It should also be noted that the data are drawn from a time of the day (noon) when the market is extremely active. Clustering is more pronounced at other times.

These features of the data are broadly consistent with a model that distinguishes between the minimum tick feasible with the markets technology and regulations and the implicit tick that arises as a convention among traders. From a data modeling perspective, we might conjecture that the tick size flips between one and five, with the former being somewhat more likely.

The model described in the next section is parameterized by the mean and variance of a quote exposure cost that approximates the “half-spread”, the variance of the daily random walk component of the price and the probability that the implicit tick size at a given time is five. Table 1 summarizes related quantities computed from the discrete data.

3. The basic model.

This section describes the basic statistical model. No claim is made that the specification is in any sense optimal. Rather, the intent is to establish a specification that is comprehensive enough to capture the features of the representative quote series discussed in the last section: the bid-ask spread, discreteness and clustering. Extensions to the model will be developed in Section 6.

The latent price variable in the market is an implicit efficient price that is economically interpreted as the expectation of end-of-trading value of the security conditional on all public information. It is assumed to follow a homoscedastic lognormal random walk:

$$\begin{aligned} m_t &= m_{t-1} + u_t \\ u_t &\sim N(0, \sigma_u^2) \end{aligned} \quad (1)$$

where m_t is the logarithm of the implicit efficient price and the u_t reflect updates to public information.

Dealers are assumed subject to a nonnegative cost of market making that impound fixed costs of their operation and per-trade costs reflecting their exposure to volatility and asymmetric information. This cost is modeled as a lognormal random variable. Letting

$$c_t \sim N(\mu_c, \sigma_c), \quad (2)$$

we assume that in the absence of tick restrictions, a market maker would bid $\exp(m_t) - \exp(c_t)$ and offer (ask) $\exp(m_t) + \exp(c_t)$.

The mapping of these continuous analogs of the quotes onto the discrete quotes is an asymmetric rounding process with a random increment. The effective tick size is a Bernoulli random variable:

$$K_t = \begin{cases} 1, & \text{with probability } k \\ 5, & \text{with probability } 1-k \end{cases} \quad (3)$$

The discrete bid and ask quotes are then given as

$$q_t = \begin{bmatrix} bid_t \\ ask_t \end{bmatrix} = \begin{bmatrix} \text{Floor}(\exp(m_t) - \exp(c_t), K_t) \\ \text{Ceiling}(\exp(m_t) + \exp(c_t), K_t) \end{bmatrix} \quad (4)$$

where the floor and ceiling functions round down (or up, respectively) to the next multiple of K_t .

When the effective tick size is unity, the mapping from latent continuous variables to the posted discrete spreads given by equations (3) and (4) may be justified by models of Bertrand competition in which no competitor is willing to make a market at an expected loss. This view is compatible with most models that incorporate discreteness (Glosten (1994), Chordia and Subrahmanyam (1995) and Kandel and Marx (1997)). On the other hand, Bernhardt and Hughson (1996) show that in a market where absolute time priority is enforced (first-come, first-

served at a given price), strategic considerations may cause dealers to post bid and ask quotes that are less aggressive than the present rounding model would suggest. Time priority is not observed, however, in most dealer markets (such as the FX market). Even in equity markets time priority is enforced at best locally.

The underlying economic rationale supporting a large implicit tick convention (five ticks, in this application) can be drawn from the models of Dutta and Madhavan (1997) and Kandel and Marx (1997). Although these analyses were constructed to model the pre-reform NASDAQ market, many of the same institutional features are present in the FX market as well. The FX markets and pre-reform NASDAQ are/were both dealer markets with non-anonymous quotes and a minimum tick size. Both markets possess/possessed an interdealer market in which dealers could lay off positions at prices that are not available to outside customers.⁶ In both markets there are/were arrangements or affinities that bind individual customers to particular dealers.⁷

For a rogue dealer who is contemplating breaking the established implicit tick convention, both of these features decrease the expected benefits and increase the expected costs. The strong relationships between customers and their dealers suggest that more aggressive pricing will not be rewarded by increased market share. If the aggressive pricing leads to the dealer being shunned in the interdealer market, their inventory management costs will increase.

Similarities notwithstanding, there are also significant differences between the two markets. While the quotes set by pre-reform NASDAQ dealers established actual transaction prices for a large volume of retail trades, customers in the FX market are primarily institutions who have more leverage in negotiation. As noted in Section 2, the FX market comprises

⁶ NSADAQ had an interdealer system (SelectNet) on which dealers could post quotes visible only to other dealers. The FX market, too, has interdealer brokers. The institution seems to be a regular feature of dealer markets, being present as well in the U.S. government bond market and the London Stock Exchange (as of this writing, prior to the impending October, 1997 reforms).

⁷ The NASDAQ market features preferencing and payment for order flow. In the FX market, it is the usual practice for a customer to check quotes only at banks with whom they already have a relationship of some sort.

alternative trading mechanisms that may offer bids and asks that are better than the indicative quotes used here. Thus, transaction prices (which are not reported in the FX market) may reflect an effective spread that is substantially narrower than that publicly quoted. Furthermore, the clustering behavior on the pre-reform NASDAQ market was a fairly stable phenomenon. Transitions from eighth quoting to quarter quoting and vice versa were rare. In contrast, clustering in the DM/\$ market is much more transient, with no obvious regime breaks.

The observed data consist of discrete bid and offer quotes. From a purely descriptive viewpoint, one approach would be to model these transitions directly using the ordered probit approach implemented by Hausman, Lo and MacKinlay (1992). (Bollerslev and Melvin (1994) estimate an ordered probit model for the spread.) In this context, an ordered probit model is a reduced-form specification, capable of modeling the spread or quote transitions in a highly flexible fashion. By way of contrast, the mapping from latent continuous to observable discrete quantities in the present model reflects a more developed (and more restrictive) economic structure. As is usually the case, choice of approach turns on whether the objective is fitting/forecasting the observed data, or investigating latent economic features of interest.

4. Estimation

a. Overview

The model for the evolution of the unobserved state variables $s_t = \{m_t, c_t, K_t\}$ is given by equations (1)-(3). The mapping from state variables to discrete quotes is given by equation (4). We observe a sample of bid and ask quotes $q = \{q_1, q_2, \dots, q_T\}$ and wish to make inferences concerning the model parameters $\Theta = \{\sigma_u^2, \mu_c, \sigma_c^2, k\}$.

Despite the model's overall simplicity, it is nonlinear in the mapping of state variables to observations and is at least partially non-Gaussian. Estimation in the traditional fashion involves computing a likelihood function using numerical integration (Kitagawa (1987)). The conditional density $f(s_t | q_t, q_{t-1}, \dots, q_1)$ is approximated by a step function on a grid defined over the state variables. The update and computation of $f(q_{t+1} | q_t, q_{t-1}, \dots, q_1)$ involve numerical integration of $f(s_{t+1} | s_t) f(s_t | q_t, q_{t-1}, \dots, q_1)$. Accuracy requires a fine grid (many points); the update at each t requires as many numerical integrations as there are points in the grid; the computation must be

performed for each t ; and of course, the optimization process itself requires many evaluations of the likelihood function. Hasbrouck (1997) describes such a calculation for a related model.

As an alternative, the present paper employs a Bayesian Markov chain Monte Carlo (MCMC) approach. Given a prior distribution on the parameter set $f(\Theta)$, we seek a posterior $f(\Theta|q)$ to use in further analysis or tabulation. This posterior is constructed as the marginal of the joint distribution $f(\Theta, s|q)$ where $s = \{s_1, s_2, \dots, s_T\}$ is the set of unobserved states. The joint distribution is in turn constructed as a Gibbs sampler that iterates over two conditional distributions. Specifically:

Step 0. Initialize s .

Step 1. Draw new Θ from $f(\Theta|s, q) = f(\Theta|s)$.

Step 2. Draw new s from $f(s|\Theta, q)$

Step 3. Repeat steps 1 and 2.

Each iteration of this process is called a scan or sweep. Under mild regularity conditions, in the limit (as the number of iterations goes to infinity), successive draws of s and Θ can be viewed as (dependent) draws from the target distribution $f(\Theta, s|q)$. The above algorithm describes the implementation of the Gibbs sampler at the “top level” of the problem, successively conditioning on Θ and s . In what follows, Gibbs samplers are also implemented within steps 1 and 2, to draw from $f(\Theta|s)$ and $f(s|\Theta, q)$.

b. Sampling from $f(\Theta|s)$

The distribution $f(\Theta|s)$ is proportional to the product $f(s|\Theta)g(\Theta)$, where $f(s|\Theta)$ is the usual model likelihood and $g(\Theta)$ is the prior distribution. (Were the states observed, $f(\Theta|s)$ would be the posterior parameter distribution.) Although $f(\Theta|s)$ possesses no convenient closed form, random draws from this distribution can be generated using a Gibbs sampler. That is, we may make sequential draws from $f(\theta_i|s, \Theta_{/\theta_i})$ where $\Theta_{/\theta_i} = \{\theta_j : i \neq j\}$.

For convenience we assume noninformative priors that are independent across parameters. The following material summarizes standard textbook results from Bayesian statistical analysis (e.g., Carlin and Louis (1996), Press (1989) or Tanner (1996)). The noninformative prior for μ_c is flat over the real line (formally, $f(\mu_c) \propto \kappa$ for an arbitrarily small κ). The posterior is conditional on T (simulated) observations of c_t , since $c_t \in s_t$, and also on

$\sigma_c^2 \in \Theta_{/\mu_c}$. The posterior $f(\mu_c | s, \Theta_{/\mu_c})$ is normal with mean $\bar{c} = (\sum c_t)/T$ and variance σ_c^2/T . The noninformative prior for σ_c^2 is $f(\sigma_c^2) \propto 1/\sigma_c^2$. The posterior is $(\sum \tilde{c}_t^2) \chi_{df=T}^{-2}$ where $\tilde{c}_t = c_t - \mu_c$. Similarly for σ_u^2 , with a prior $f(\sigma_u^2) \propto 1/\sigma_u^2$, the posterior is $(\sum \tilde{u}_t^2) \chi_{df=T-1}^{-2}$ where $\tilde{u}_t = m_t - m_{t-1}$. The noninformative prior for k is the beta distribution $B(\alpha = 1/2, \beta = 1/2)$. The posterior is $B(\alpha = 1/2 + n, \beta = 1/2 + T - n)$ where n is the number of occurrences of “1” in the set $K = \{K_1, K_2, \dots, K_T\}$.

c. *Sampling from $f(s|\Theta, q)$.*

We may sample from $f(s|\Theta, q)$ by successively drawing from $f(s_t | s_{/t}, \Theta, q)$ where $s_{/t} = \{s_i : i = 1, \dots, T \text{ and } i \neq t\}$, i.e., a Gibbs sampler over t . At each time t , furthermore, we may draw from $f(s_t | s_{/t}, \Theta, q)$ by successively drawing from the components of s_t , from $f(c_t | m_t, K_t, s_{/t}, \Theta, q)$, $f(m_t | c_t, K_t, s_{/t}, \Theta, q)$ and $f(K_t | m_t, c_t, s_{/t}, \Theta, q)$, i.e., a Gibbs sampler within each t .

To sample from $f(c_t | m_t, K_t, s_{/t}, \Theta, q) = f(c_t | m_t, K_t, \mu_c, \sigma_c^2, q_t)$, note first that $f(c_t | \mu_c, \sigma_c^2)$ is normal. The effect of conditioning on the other variables is to enforce the restriction that the unobserved state variables map to the observed quotes (equation (4)). This mapping implies

$$\begin{aligned} bid_t &\leq \exp(m_t) - \exp(c_t) < bid_t + K_t \\ ask_t - K_t &< \exp(m_t) + \exp(c_t) < ask_t \end{aligned} \quad (5)$$

which implies

$$\begin{aligned} \text{Log}(\text{Max}[\exp(m_t) - bid_t - K_t, ask_t - K_t - \exp(m_t)]) &< c_t \\ c_t &< \text{Log}(\text{Min}[\exp(m_t) - bid_t, ask_t - \exp(m_t)]) \end{aligned} \quad (6)$$

We may therefore sample c_t from a normal density with mean μ_c and variance σ_c^2 subject to the truncation given in (6).

To sample from $f(m_t | c_t, K_t, s_{/t}, \Theta, q) = f(m_t | c_t, K_t, m_{t-1}, m_{t+1}, \sigma_u^2, q_t)$ for $t=2, \dots, T-1$, note that $f(m_t | m_{t-1}, m_{t+1}, \sigma_u^2)$ is normal with mean $(m_{t-1} + m_{t+1})/2$ and variance $\sigma_u^2/2$. Rearranging restriction (5) to isolate m_t gives:

$$\begin{aligned} \text{Log}(\text{Max}[bid_t + \exp(c_t), ask_t - K_t - \exp(c_t)]) &< m_t \\ m_t &< \text{Log}(\text{Min}[bid_t + \exp(c_t) + K_t, ask_t - \exp(c_t)]) \end{aligned} \quad (7)$$

The required draw is therefore from a truncated normal density. (The modification to the endpoints $t=1$ and $t=T$ is direct.)

To sample from $f(K_t|m_t, c_t, s_{/t}, \Theta, q) = f(K_t|m_t, c_t, k, q_t)$, note that a straightforward application of Bayes Law gives the conditional probability

$$\Pr(K_t = 1|m_t, c_t, k, q_t) = \frac{k \Pr(q_t|m_t, c_t, K_t = 1)}{k \Pr(q_t|m_t, c_t, K_t = 1) + (1-k) \Pr(q_t|m_t, c_t, K_t = 5)} \quad (8)$$

Given that the mapping from state variables to quotes is nonstochastic, the probabilities on the r.h.s. are merely indicator variables that are equal to one if the conditioning state variables are compatible with the observed quotes and zero otherwise. The draw of K_t can then be made as a Bernoulli random variable with the conditional probability given in (8).

d. Summary

It should be apparent from the preceding that the Gibbs approach thoroughly pervades this application. The Gibbs sampler is used at the “top level” of the problem to alternately draw from parameters and states. Gibbs samplers are also used to draw from the parameter vector one at a time, to draw from the states one t at a time, and (within each t) to draw state variables one at a time. It is important to note that these latter “lower level” samplers need not be iterated to convergence within each of the steps 1 and 2. Rather the entire process can be viewed as a single Gibbs sampler with many steps, each of which involves a single draw of one parameter or latent variable conditional on all the remaining parameters and variables.

The actual sequencing of the Gibbs sampler used here deviates slightly from that described above. Since the draws for c_t and K_t do not depend on the state variables at any other time, computational efficiency is enhanced by separately drawing c_t and K_t for $t=1, \dots, T$ in two (vectorized) steps. The draw of m_t , on the other hand, depends on m_{t-1} and m_{t+1} , however, and we must therefore iterate over t .

Estimation proceeds by initializing the state variables and parameters. (They may be initialized to any feasible values.) A large number of iterations are made, and the simulated values are monitored to assess the convergence. When the stream of simulated values is judged

to be sufficiently large and well mixed, posterior parameter distributions may be constructed from the simulated parameter draws.

5. Estimation of the \$/DM series.

To start the sampler, the model parameters were initialized as $\mu_c=10$, $\sigma_c^2=10$, $\sigma_u^2=10$ and $k=1$. The choice of k is convenient because it forces all K_t to be equal to one, a value that is compatible with all observed quotes. The other initializations were chosen (after preliminary testing of the sampler) as “bad” starting values, i.e. values sufficiently removed from the final parameter estimates that convergence might be challenged. The m_t were all initialized to the log midpoint of the observed bid and ask. All of the K_t were initialized to one. I computed 11,000 iterations of the sampler. The program was implemented in the matrix language OX (Doornik (1996)) and took roughly seven minutes on a 200 MHz Pentium Pro.

Figure 3 plots for each parameter the first 2,000 draws of the sampler, as well as correlograms and histograms for the last 10,000 draws (i.e., after discarding the initial 1,000). Visual inspection of the draws suggests that the sampler converges quickly. The μ_c and k parameters exhibit the slowest convergence, but even here the effects of the starting values appear to die out after roughly one hundred draws. A Gibbs sampler does not produce independent draws. Since it is difficult to discern serial dependence from plots of the draws, it is useful to examine the correlograms. The autocorrelations are generally modest, with the largest arising for the clustering probability k . The rightmost graph for each parameter gives the histogram of the simulated parameters. This is an approximation to the exact small-sample posterior distribution for the parameters. The approximation is exact in the limit as the number of sweeps goes to infinity, holding the number of observations fixed.

Table 2 reports summary statistics for these simulated posteriors. At the posterior mean values, the cost parameter $c_t \sim N(\mu_c = 0.780, \sigma_c^2 = 0.195)$. Restated to level terms using the lognormal transformation, the mean and standard deviation of $\text{Exp}(c_t)$ are 2.40 ticks and 1.11 ticks, respectively. These quantities are roughly the same magnitude as the (one-tick) resolution of the discrete spread. Both figures are substantially lower than the corresponding values computed on the basis of the observed discrete spreads (cf. 6.1 and 2.3 from Table 1)

The mean estimate of σ_u^2 is quite close to the corresponding value estimated from the discrete quote midpoints (2.06×10^{-5} vs. 2.03×10^{-5}). This is a consequence of the fact that at a daily frequency, the typical sizes of quote midpoint changes are large relative to the tick size. This may not be true, of course, for shorter intervals. Finally, the mean estimate of k implies that the implicit tick size is five roughly 13% of the time. In the discrete data, the bid and ask were found to lie on five-tick multiples 23% of the time. From the perspective of the model, however, these events include states where the implicit tick size is one, but the pre-rounded quantities just happen to lie close to a five-tick multiple.

The standard error of the mean estimate in Table 2 is a spectral estimate that corrects for autocorrelation. For example, the correlograms suggest that the most pronounced autocorrelation is found in the estimates of the k parameter. With a sample standard deviation of 0.027 and 10,000 observations (11,000 draws less the 1,000 discarded initial draws), the usual standard error of the mean under the independence assumption would be $0.027/\sqrt{10,000} = .00027$. The spectral estimate (0.00069) is roughly twice times as large.

6. Extensions

The discussion in this section turns to refinements that may make the simple model implemented here more useful and appealing. In general, the easiest extensions involve deterministic (e.g., time-of-day or –week) patterns in the parameters. These will generally require minimal modifications to the parameter and state simulations. It is also relatively easy incorporate additional sources of randomness that are serially independent. These will usually involve only an additional step in the Gibbs sampler.

Extensions that incorporate serial dependencies in the state variables are more challenging. If these dependencies can be modeled as transitions of Gaussian state variables, Shephard (1994) among others suggests that simulation is still quite practical. It is important to emphasize that the model need not be fully Gaussian, only conditionally so. The framework for dealing with limited dependent variable processes suggested by Manrique (1997) and Manrique and Shephard (1997a, b) is particularly useful in this respect.

Perhaps most importantly, the MCMC approach does not suffer from the curse of dimensionality. Incorporation of additional state variables or parameters requires only that these quantities be simulated in each sweep. The incremental computational burden of this simulation does not (to a first approximation) depend on the number of variables or parameters already present in the model.

a. Time-varying volatility

The approach described to this point can easily accommodate volatility that changes in a deterministic fashion. Section 4.c described a homoscedastic scan sampler for $f(s|\Theta, q)$. This may be generalized by replacing σ_u^2 in equation (1) with $\sigma_{u,t}^2$ where $\sigma_{u,t}^2 = \text{Var}(u_t)$. The resulting scan sampler is valid conditional on any sequence of variances $\{\sigma_{u,1}^2, \sigma_{u,2}^2, \dots, \sigma_{u,T}^2\}$, provided that the u_t are independent of the variance sequence. We will call this modification the "independent variance scan sampler". We might assume, for example, that the variance takes on one of two possible values depending on whether the interval from $t-1$ to t spans a weekend. We may posit two noninformative priors, and update these priors using only the relevant observations. This approach can be used to handle deterministic intraday and weekday volatility patterns commonly found in microstructure data.

Volatility that changes in a random fashion is a little more involved. There are two main approaches here. In ARCH-family models, $\sigma_{u,t}^2$ depends solely on prior disturbances u_t . In stochastic volatility models, $\sigma_{u,t}^2$ follows a random process with increments that are independent of u_t at all leads and lags. ARCH-type models predominate in the empirical literature, probably in part due to the ease of estimation. Stochastic volatility models predominate in the MCMC literature, however, at least in part for the reason that MCMC approaches provide the only practical approach to estimation. Either ARCH or stochastic volatility extensions could be implemented here. These are obvious and desirable refinements.

b. Time-varying and serially correlated trading costs

In many markets discrete spreads are found to exhibit periodicities at daily frequencies. Spreads are typically elevated at the start and end of trading sessions. The present framework can easily accommodate such deterministic variation by allowing one or both of the quote exposure cost parameters (μ_c and/or σ_c^2) to be a function of time-of-day. It is also reasonable to

suggest that the cost might have persistent stochastic components arising from serial correlation in some of the cost determinants (such as volatility and asymmetric information). Hasbrouck (1997) suggests an autoregressive model.

c. Asymmetric bid and ask costs

In the present model, the quote exposure cost is the same on both the bid and ask sides of the market. This is a reasonable assumption if the same quote-setter is active on both sides of the market, and if the quotes are not sensitive to current inventory positions. In many markets, however, the bid and ask quotes may reflect different individual limit order traders, presumably subject to different trading costs. Hasbrouck (1997) captures this by modeling trading costs as two processes that evolve independently of each other.

d. Alternative clustering models

The implicit tick size in the simple model is a Bernoulli random variable. In some applications, a more refined multinomial specification may be desirable. Harris (1991), for example, suggests that the implicit tick size may be a function of volatility and trading costs.

7. Conclusions

Structural models of security prices over short intervals usually reflect microstructure effects arising from the manner in which the security is traded. This paper explores a model in which a latent continuous efficient price and latent continuous trading costs map to bid and ask quotes that are discrete and clustered. Despite the simplicity of the model, there are no convenient closed-form representations for moments or conditional probabilities. Inference using minimum distance estimators (maximum likelihood or method of moments) is therefore beset with computational difficulties. In a Bayesian framework, however, inference is greatly simplified by the application of simulation (Markov chain Monte Carlo) methods. An application to daily \$/DM exchange rates illustrates the usefulness of the method.

8. References

- Ahn, H.-J, C. Q. Cao, and H. Choe, 1996, Tick size, spread and volume, *Journal of Financial Intermediation*, 5.
- Angel, J., 1994, Tick size, share prices and stock splits, working paper, Georgetown University.
- Anshuman, V. R., and Avner Kalay, 1994, Market making rents under discrete prices: Theory and evidence, Working paper, Boston College.
- Ball, C. A., 1988, Estimation bias introduced by discrete security prices, *Journal of Finance*, 43, 841-865.
- Bernhardt, D. and E. Hughson, 1996, Discrete pricing and the design of dealership markets, *Journal of Economic Theory*, 71, 148-182.
- Bessembinder, H., Bid-ask spreads in the interbank foreign exchange markets, *Journal of Financial Economics*, 35, 317-348.
- Bollerslev, T., and M. Melvin, 1994, Bid-ask spreads in the foreign exchange market: an empirical analysis, *Journal of International Economics*, 36, 355-372.
- Brown, S., P. Laux and B. Schacter, 1991, On the existence of an optimal tick size, *Review of Futures Markets*, 10, 50-72.
- Carlin, B. P. and T. A. Louis, 1996, *Bayes and Empirical Bayes Methods for Data Analysis* (Chapman and Hall, London).
- Carter, C. K. and R. Kohn, 1994, Gibbs sampling for state space models, *Biometrika*, 81, 541-53.
- Casella, G. and E. I. George, 1992, Explaining the Gibbs sampler, *American Statistician*, 46, 167-190.
- Chib, S. and E. Greenberg, 1996, Markov Chain Monte Carlo simulation methods in econometrics, *Econometric Theory*, 12, 409-431.
- Chib, S. and E. Greenberg, 1995, Understanding the Metropolis-Hastings algorithm, *American Statistician*, 49, 327-335.
- Chordia, T. and A. Subrahmanyam, 1995, Market making, the tick size, and payment-for-order flow: theory and evidence, *Journal of Business*, 68, 543-575.
- Christie, W. G. and P. H. Schultz, 1994a, Why do NASDAQ market makers avoid odd-eighth quotes?, *Journal of Finance*, 49, 1813-40.
- Christie, W. G. and P. H. Schultz, 1994b, Why did NASDAQ market makers stop avoiding odd-eighth quotes?, *Journal of Finance*, 49, 1841-60.
- Cordella, Tito and Thierry Foucault, 1996, Minimum price variations, time priority and quote dynamics, Working paper, Universitat Pompeu Fabra.
- de Jong, P. and N. Shephard (1995) The simulation smoother for time series models, *Biometrika*, 82, 339-50.

- Dravid, A. R., 1991, Effects of bid-ask spreads and price discreteness on stock returns, Working Paper, Rodney L. White Center, Wharton School.
- De Jong, P., 1991, The diffuse Kalman filter, *Annals of Statistics*, 19, 1073-1083.
- Doornik, J.A., 1996, Object-Oriented Matrix Programming using Ox, London: International Thomson Business Press and Oxford: <http://www.nuff.ox.ac.uk/Users/Doornik/>.
- Dutta, P. K. and Madhavan, A., 1997, Competition and collusion in dealer markets, *Journal of Finance*, 52, 245-276.
- Gilks, W. R., S. Richardson and D. J. Spiegelhalter, 1996, Introducing Markov chain Monte Carlo, in W. R. Gilks, S. Richardson and D. J. Spiegelhalter, eds., *Markov Chain Monte Carlo in Practice* (Chapman and Hall, London).
- Glosten, L. R., 1994, Is the electronic open limit order book inevitable?, *Journal of Finance*, 49, 1127-1161.
- Glosten, L. R. and L. E. Harris, 1988, Estimating the components of the bid-ask spread, *Journal of Financial Economics*, 21, 123-42.
- Gottlieb, G. and A. Kalay, 1985, Implications of the discreteness of observed stock prices, *Journal of Finance*, 40, 135-153.
- Harris, L. E., 1990, Estimation of stock variances and serial covariances from discrete observations, *Journal of Financial and Quantitative Analysis*, 25, 291-306.
- Harris, L. E., 1991, Stock price clustering and discreteness, *Review of Financial Studies*, 4, 389-415.
- Harris, L. E., 1994, Minimum price variations, discrete bid-ask spreads and quotation sizes, *Review of Financial Studies*, 7, 149-178.
- Harris, L. E., 1997, Decimalization: a review of the arguments and evidence, working paper, School of Business, University of Southern California.
- Hasbrouck, J., 1996, Modeling microstructure time series, in *Handbook of Statistics, Volume IV*, G. S. Maddala and C. R. Rao, eds., (North Holland; Amsterdam).
- Hasbrouck, J., 1997, The dynamics of discrete bid and ask quotes, Working paper, Stern School, New York University.
- Hausman, J. A., A. W. Lo and A. C. MacKinlay, 1992, "An Ordered Probit Analysis of Transaction Stock Prices," *Journal of Financial Economics*, 31, 319-330.
- Jacquier, E., N. G. Polson and P. E. Rossi, 1994, Bayesian analysis of stochastic volatility models, *Journal of Business and Economic Statistics*, 12, 371-389.
- Kandel, E. and L. M. Marx, 1997, Nasdaq market structure and spread patterns, *Journal of Financial Economics*, 45, 61-89.
- Kim, S., N. Shephard and S. Chib, 1996, Stochastic volatility: likelihood inference and comparison with ARCH models, *Review of Economic Studies*, forthcoming.
- Kitagawa, G., 1987, "Non-Gaussian State-Space Modeling of Nonstationary Time Series," *Journal of the American Statistical Association*, 82, 1032-1041.

- Lyons, R., 1995, Tests of microstructural hypotheses in the foreign exchange market, *Journal of Financial Economics*, 39, 321-351.
- Manrique, A., 1997, *Econometric Analysis of Limited Dependent Variable Time Series*, Ph.D. Thesis, Nuffield College, Oxford University.
- Manrique, A. and N. Shephard, 1997a, Simulation based likelihood inference for limited dependent variable processes, Working paper, Nuffield College, Oxford University.
- Manrique, A. and N. Shephard, 1997b, Likelihood analysis of a discrete bid/ask price model for a common stock quoted on the NYSE, Working paper, Nuffield College, Oxford University
- O'Hara, M., 1995, *Market Microstructure Theory*, (Blackwell Publishers; Cambridge, MA).
- Peiers, B., 1997, Informed traders, intervention and price leadership: a deeper view of the microstructure of the foreign exchange market, *Journal of Finance*, 52, 1589-1614.
- Press, S.J., 1989, *Bayesian Statistics: Principles, Models and Applications*, (John Wiley, New York).
- Schwert, G. W., 1997, Symposium on market microstructure: Focus on Nasdaq, *Journal of Financial Economics*, 45, 1-8.
- Shephard, N., 1994, Partial non-Gaussian state space, *Biometrika*, 81, 115-131.
- Shephard, N. and A. Manrique, 1997, Simulation based likelihood inference for limited dependent processes, Working paper, Nuffield College, Oxford University.
- Shephard, N. and M. K. Pitt, 1997, Likelihood analysis of non-Gaussian measurement time series, *Biometrika*, 83, 653-67.
- Smith, A. F. M. and A. E. Gelfand, 1992, Bayesian statistics without tears: a sampling-resampling perspective, *American Statistician*, 46, 84-88.
- Tanner, M. A., 1996, *Tools for Statistical Inference*, (Springer-Verlag, New York).
- U.S. Securities and Exchange Commission, 1996, Report Pursuant to Section 21(a) of the Securities Exchange Act of 1934 Regarding the NASD and the NASDAQ Market, (available at www.sec.gov/news/extra/nasd2.htm).
- Yao, J., 1997, Market making in the interbank foreign exchange market, Working paper, Stern School of Business, New York University.

Table 1.
Preliminary Descriptive Statistics for \$/DM Quotes

Mean of bid-ask spread (ticks)	6.1
Standard deviation of bid-ask spread (ticks)	2.3
Variance of daily log quote midpoint price changes	2.03×10^{-5}
Percentage of quotes in which both the bid and ask lie on a five-tick multiple.	23.4%

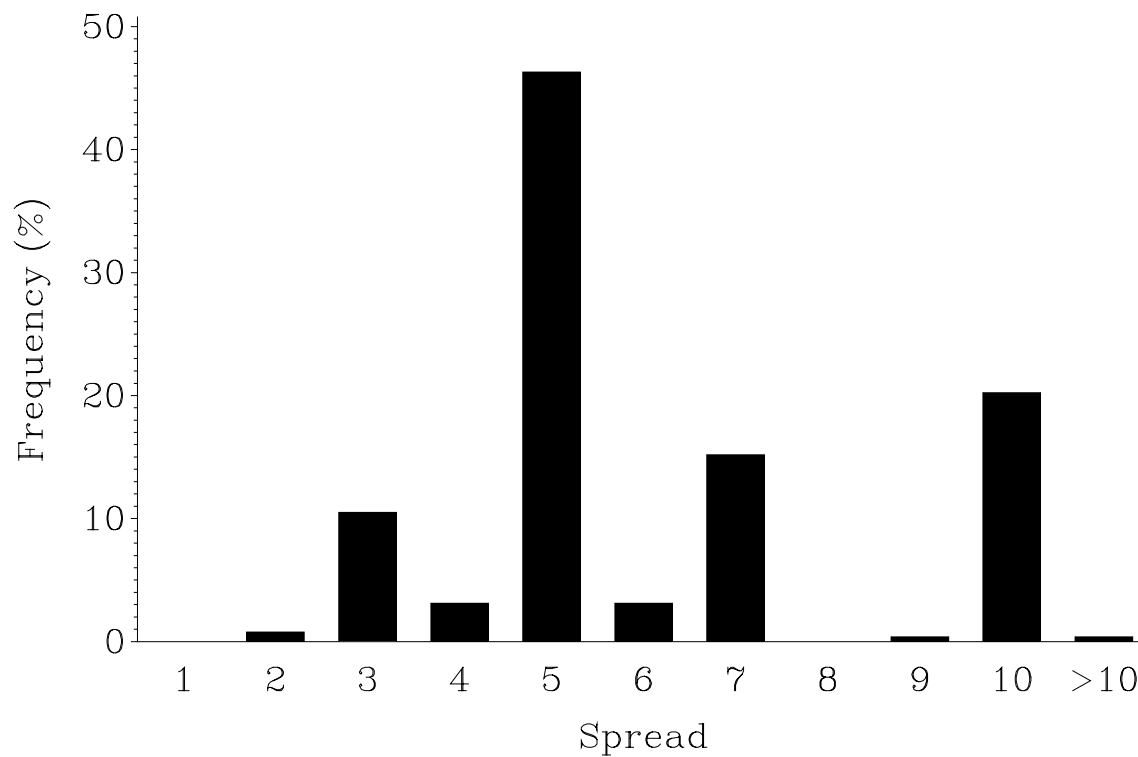
Notes: \$/DM quotes as of noon (GMT) for all trading days in 1996. (Source: Olsen Associates HFDF II).

Table 2.
Summary of Posterior Parameter Distributions

Parameter	Mean	Std.Error of Mean	Std. Dev.
μ_c	0.780	0.00046	0.030
σ_c^2	0.195	0.00040	0.021
σ_u^2	2.06×10^{-5}	1.8×10^{-8}	1.8×10^{-6}
k	0.874	0.00069	0.027

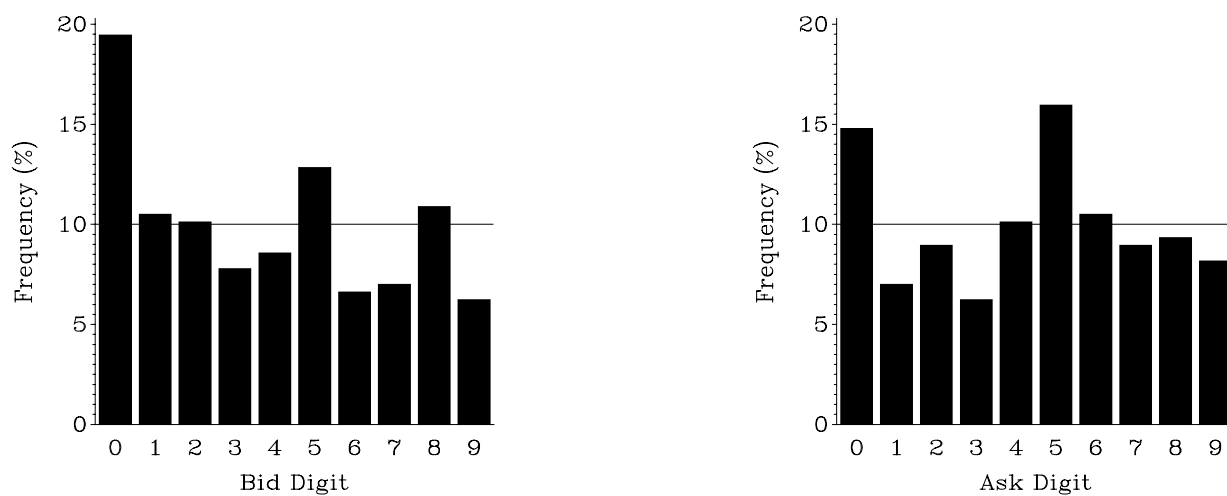
Notes: Summary statistics for simulated posterior distributions constructed using a Gibbs sampler over 11,000 iterations, with the first 1,000 discarded. The standard error of the mean estimates is a spectral estimate (using a Parzen window) that corrects for autocorrelation in the sample draws.

Figure 1
Distribution of Bid-Ask Spread



Notes: Frequency (in DM 0.0001 ticks) of bid ask spreads for \$/DM exchange rates as of noon, GMT, 1996

Figure 2
Distributions of Last Digits in Bid and Ask Quotes



Notes: Frequency (in DM 0.0001 ticks) of the last digit in the bid and ask quotes for all trading days in 1996 as of noon GMT.

Figure 3. Simulated Parameters.

Figure is based on a single Gibbs sampler run of 11,000 iterations as described in Section 4. For each of the model parameters, the leftmost graph depicts the first 1,000 draws. The middle graph shows the autocorrelations of the last 10,000 draws (after the first 1,000 were discarded as burn-in period). The rightmost graph is the histogram of the last 10,000 draws.

